

Journal of Herpetology, Vol. 25, No. 2, pp. 226-229, 1991
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**Satellite Behavior in the Neotropical
Treefrog *Hyla minuta***

CÉLIO F. B. HADDAD, *Departamento de Zoologia, Instituto
de Biociências, UNESP, Caixa Postal 178, 13500 Rio Claro,
São Paulo, Brasil*

Individuals of the same sex may employ alternative
mating tactics to increase their reproductive success



FIG. 1. Satellite male association in the neotropical treefrog *Hyla minuta*. The arrow indicates the satellite male.

(Waltz, 1982). In several anurans, non-calling satellite males have been observed associated with calling males (e.g., Wells, 1977a; Arak, 1988). Wells (1977b) proposed two hypotheses to explain satellite behavior in anurans: (1) satellite males are waiting for call site vacancies; or (2) satellite males are sexual parasites attempting to intercept gravid females approaching calling males. These hypotheses need not be mutually exclusive. I studied the satellite behavior of a neotropical treefrog, *Hyla minuta*, to test these hypotheses.

The study site was a small artificial pond (27 m shoreline) near Campinas (22°48'S; 47°05'W), São Paulo State, southeastern Brazil. A total of 47 breeding choruses (comprised of at least two calling males) were monitored from May 1984 to April 1985. Additional observations were made in 1983 and 1989. Observations and field experiments were conducted between 1800 and 0100 h using a dim or red flashlight. I censused breeding choruses by walking around the edges of the pond and recording the position, mating behavior, and identity of individuals. Some, but not all, males were individually recognized by means of natural markings. A male was classified as a satellite based on its distinctive low posture, non-calling behavior, and proximity to a calling male (within 1 m).

To test the two hypotheses for satellite behavior, I removed the calling male and left the satellite in place. This procedure was replicated 10 times; in each case the calling male and its satellite were at least 1 m from the next nearest calling male. I recorded the number of satellites that started to call or that established new satellite associations. To better examine the parasite hypothesis, I captured 10 females in am-

plexus and separated them from their mates. Females were cooled to 7°C to inhibit oviposition in captivity and held for one or two days. These females were then released at the breeding site near calling-satellite pairs, equidistant from the males. I scored the mating success of males exhibiting the two alternative behaviors. The snout-vent lengths (SVLs) of all males were measured in the field with calipers to the nearest 0.1 mm.

Approximately 83.9% (47/56) of the satellite associations of *H. minuta* were composed of one calling male and one satellite; 10.7% (6/56) were composed of one calling male and two satellites; 3.6% (2/56) were composed of two calling males and one satellite (the satellite was equidistant between two calling males); and 1.8% (1/56) consisted of one calling male and three satellites. The ratio of satellite to calling males was not correlated with the number of calling males (Kendall's $\tau = 0.152$, $P = 0.174$, $N = 47$ nights), which indicates that the proportion of satellites did not increase as a function density. This corroborates data presented for *Hyla picta* and *H. chrysoscelis* (Roble, 1985), but contrasts with data for *Bufo cognatus* (Sullivan, 1982). As expected, the number of satellite males was correlated with the number of calling males ($\tau = 0.332$, $P = 0.004$, $N = 47$ nights).

The distance between neighboring calling males ($\bar{x} = 130.16$ cm, $SD = 118.68$, $N = 158$) was greater ($t = 11.11$; $P < 0.01$) than the distance between calling and satellite males ($\bar{x} = 20.90$ cm, $SD = 18.77$, $N = 47$). Satellite males adopted a crouched posture in close proximity to calling males, with their lungs and vocal pouches deflated (Fig. 1). Calling males ($\bar{x} = 26.2$ mm, $SD = 1.00$, $N = 26$) were significantly larger in SVL

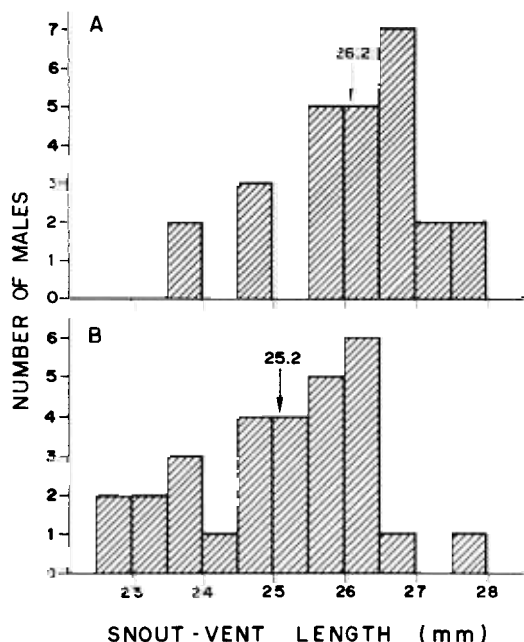


FIG. 2. Distributions of body sizes (SVL) for calling (A) and satellite (B) males of *Hyla minuta*. Arrows indicate the median sizes.

($t = 3.15$, $P < 0.01$) than satellites ($\bar{x} = 25.2$ mm, $SD = 1.23$, $N = 29$; Fig. 2). Satellites are smaller than calling conspecifics in many species of anurans in which they have been observed, suggesting that the satellites may be younger or less dominant males (Howard, 1978; Forester and Lykens, 1986; Robertson, 1986; Given, 1988). Consistent with this, satellite males in *H. minuta* were commonly observed to be the losers in territorial contests. Active fighting and exchanges of encounter calls appeared to determine the tactic (calling or satellite) adopted by males after contests ($N = 47$). However, a satellite association was established in some instances without any noticeable aggressive behavior by the calling male ($N = 7$); in few instances, encounter calls without fighting were enough to induce satellite behavior ($N = 3$).

Individual males of *H. minuta* switched between satellite and calling behavior within the same night, and between successive nights. Satellite associations lasted from a few minutes to about 6 h (the entire calling period in a given night). When the calling male did not achieve amplexus, the association was maintained for the entire calling period ($N = 6$); however, when the calling male achieved amplexus or moved to another location, the associated satellite started to call ($N = 4$). Switching between satellite and calling behavior, similar to that observed in *H. minuta*, has been reported for other anurans (Howard, 1978; Perrill et al., 1978, 1982; Mac Nally, 1979; Perrill 1984; Roble, 1985; Arak, 1988; Krupa, 1989). Arak (1988) suggested that a shift between satellite and calling behavior may be partly explained by "local effects": when a male is close to an unattractive male, it may be better to call to attract females; whereas when a

male is close to an attractive male, it may be better to change to satellite behavior.

When the calling male of *H. minuta* was removed, in eight instances the satellite began to call after a few minutes, offering support for the "wait for call site vacancy hypothesis." In two cases the satellite moved to the closest calling male and established a new association, in agreement with the "parasite hypothesis." In four natural instances I observed a gravid female attracted to a calling male associated with a satellite. Calling males attempted amplexus in two cases and succeeded in one; satellite males attempted to intercept the four females and succeeded twice. One female fled when approached by both males simultaneously. During the field experiment that involved placement of a female near a calling-satellite pair, six females ignored the satellite association and selected other males within the chorus; the remaining four females were attracted to the calling-satellite pairs. Calling males attempted amplexus in three cases and succeeded in all; satellite males attempted to intercept the females in two cases and succeeded in one. The rate of mating success observed for satellite males in *H. minuta* (natural + staged) was identical to that observed for satellites in *Hyla ebraccata* (38%; Miyamoto and Cane, 1980), and similar to the rate observed for satellites in *Hyla cinerea* (43%; Perrill et al., 1978). The field experiments indicate a dual role for satellite behavior in *H. minuta*. In some instances males appear to be waiting for call site vacancies; in others they are acting as sexual parasites. Both modes presumably facilitate access to females, and are not mutually exclusive.

Acknowledgments.—I thank A. S. Abe, A. J. Cardoso, T. M. Lewinsohn, M. Martins, S. F. Reis, I. Sazima, K. D. Wells, two anonymous reviewers, and B. K. Sullivan for helpful comments on earlier versions of the manuscript; I also thank Fundação MB and CAPES for financial support.

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Accepted: 25 October 1990.